INTRODUCTION TO NETWORKING

CIS*3210

Fall 2026

Sajid Marhon

Based on notes for Computer Networking: A Top-Down Approach, 9th ed., by J. Kurose and K. Ross

Learning Outcomes

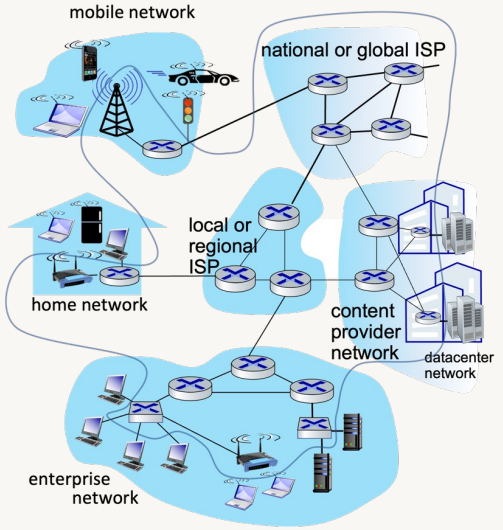


- Explain the growth and importance of computer networks
- Identify five key aspects of computer networking
- Explain protocols and protocol suites
- Describe the benefits of layering

Growth of Computer Networks



- Since the 1970s, computer networking has grown from a research topic into essential infrastructure
- Networks support many areas of society, including:
  - Business: advertising, production, shipping, planning, billing, and accounting
  - Education: access to online resources for students and instructors
  - Government: daily operations at all levels
- Computer networks are now everywhere



Growth of Computer Networks



- Today, it is a global communication system
- Networking has transformed the economy by:
  - Enabling telecommuting
  - Changing business communication
  - Creating new technologies, products, services, and industries
  - Increasing demand for networking expertise
  - Requiring skilled professionals to design, install, operate, and manage networks



1980
Research Network
Only a few dozen connected sites

Expansion
People and organizations connect worldwide

Transformation
Economic and Social Change
New industries, business communication, and telecommuting

Today
Essential Infrastructure
Growing demand for networking professionals

Growth of Computer Networks



Discussion Question:

Why can growth in network use increase both the value and the fragility of an organization?

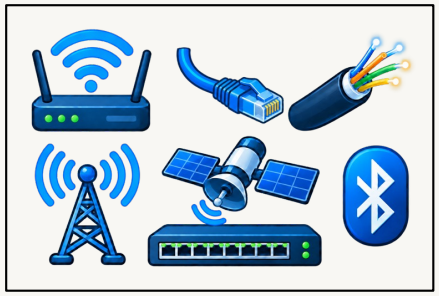
More connected users and services create coordination, reach, and network effects.

They also increase dependency, attack surface, congestion risk, and the number of shared components whose failure can affect many activities.

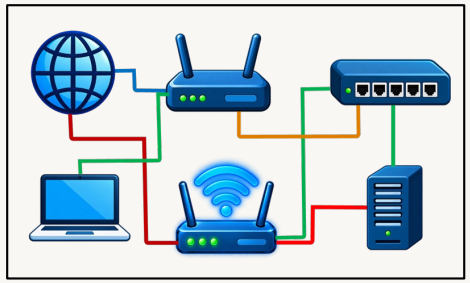
Why Networking Seems Complex



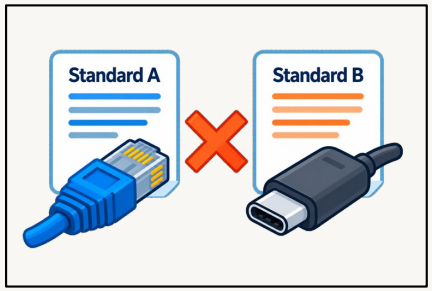
- Networking is complex because:
  - Many technologies exist, each with distinct features
  - Companies apply technologies in different and innovative ways
  - Technologies can be combined and interconnected in many ways
- Networks can be confusing for beginners because:
  - No single theory explains how all components relate
  - Standards from different organizations may be incompatible
  - Conceptual models may be too simple, incomplete, or overly complex
  - Rapid technological change makes standardization difficult



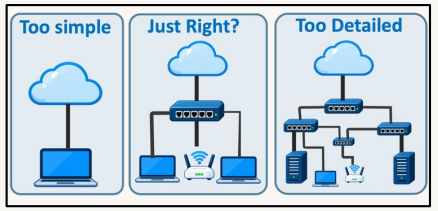
Many technologies



Many combinations



Different standards



Imperfect models

Why Networking Seems Complex



- Inconsistent terminology makes networking difficult for beginners:
  - Different groups use their own terms
  - Researchers prefer precise scientific language
  - Marketing teams create labels to distinguish products
  - Technical terms are often confused with brand names
  - Terms may be reused across technologies, causing ambiguity
  - Overlapping acronyms and product-specific jargon add confusion

The Key Aspects of Networking



- A broad understanding of networking requires five key areas:
  - **Network applications:** How software communicates using sockets and APIs
  - **Data communications:** Signals, media, encoding, bandwidth, and errors
  - **Packet switching:** LANs, WANs, frames, MAC, Ethernet, and Wi-Fi
  - **TCP/IP internetworking:** Addressing, routing, TCP, UDP, and core protocols
  - **Cross-cutting topics:** Performance, security, management, IoT, and emerging technologies

Network Applications and Network Programming



- Network applications allow software on different hosts to communicate
- Examples include:
  - Email and file transfer
  - Web browsing
  - VoIP and video conferencing
  - Distributed databases
- Each application provides a distinct service and interface
- All applications share the same network infrastructure

Network Applications and Network Programming

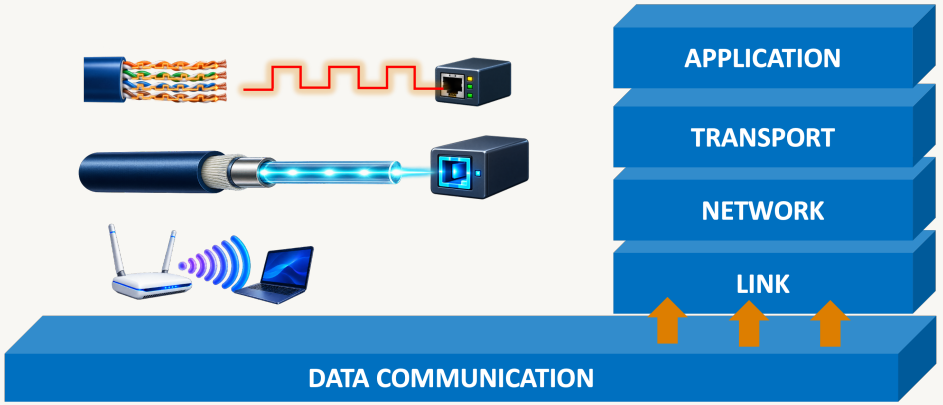


- A unified network simplifies application development
  - Developers learn one networking API and a few core functions
  - Applications can be built without knowing low-level hardware or protocols
  - Basic tasks require little additional networking knowledge
- However, understanding the network helps create faster, cleaner, and more reliable applications

Data Communications



- **Data communications** examines how information travels through wired, wireless, and optical media
- It focuses on encoding and transmitting data using physical signals
- Its principles influence many network protocol layers
- It provides the foundation for computer networking

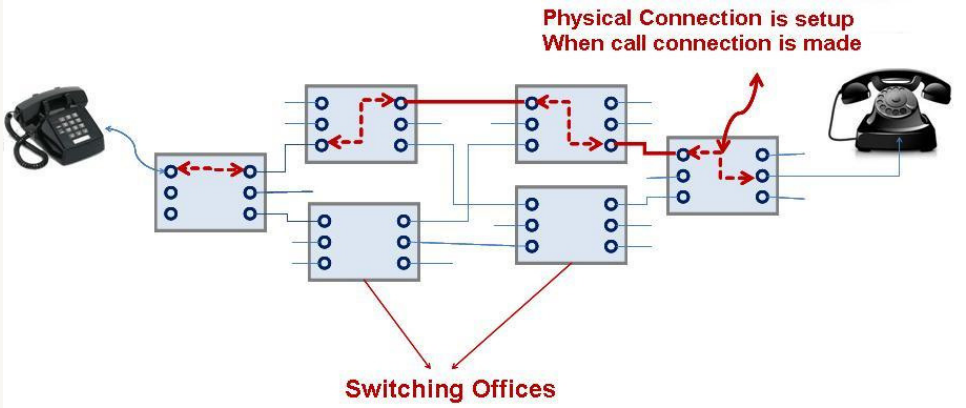


Packet Switching and Networking Technologies



- In the 1960s, **packet switching** transformed data communications
- Earlier telephone and telegraph networks used **circuit switching**, creating a dedicated end-to-end connection
- Electronic switches automated the process, but the principle remained: establish a circuit, then transmit

Circuit Switching

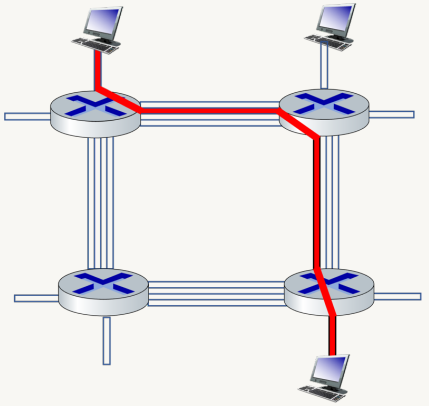


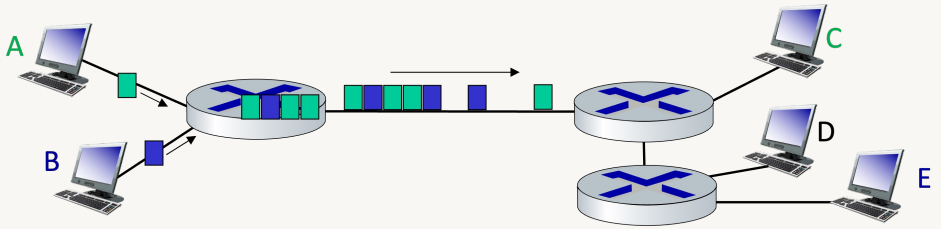
https://forum.networklessons.com/t/introduction-to-wans-wide-area-network/1443

Packet Switching and Networking Technologies



- **Packet switching** sends data without a dedicated circuit using store-and-forward transmission
- It forms the foundation of the modern Internet
- Key features:
  - Users share network resources
  - Data is divided into small packets
  - Each packet includes a destination address
  - Network devices use forwarding information to deliver packets





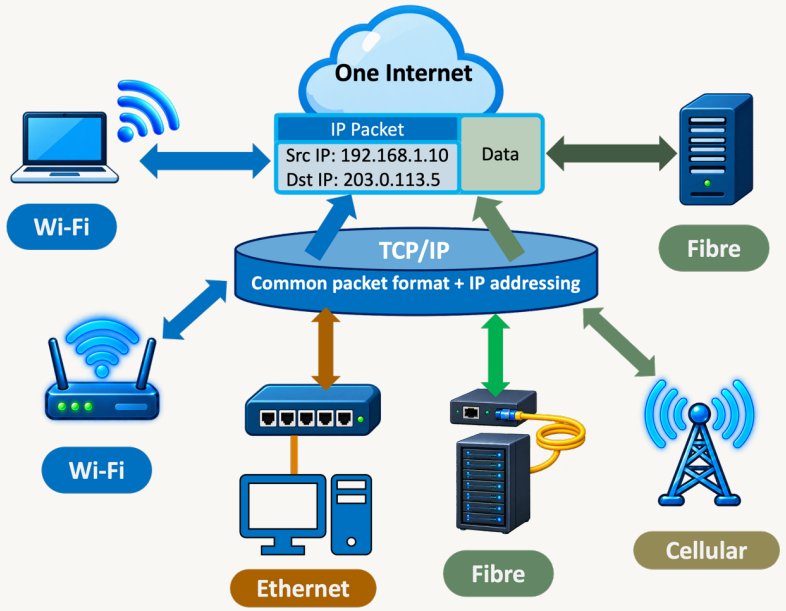
Packet Switching and Networking Technologies



- Packet-switched networks must address several design questions:
  - **Addressing:** How is the destination identified?
  - **Discovery:** How is its address found?
  - **Packet sizing and framing:** How are packets sized and separated?
  - **Medium sharing:** How do devices share a link fairly?
  - **Wireless adaptation:** How is packet switching applied to radio links?
  - **Trade-offs:** How are speed, range, and cost balanced?
- Different technologies are optimized for specific performance, distance, and cost requirements

Internetworking with TCP/IP

- In the 1970s, the Internet emerged as a new approach to networking
- No single packet-switching technology could meet every need
- The solution was to interconnect different networks using open standards
- These standards formed the **TCP/IP protocol suite**
- TCP/IP supports diverse technologies through a common packet format and addressing system





Discussion Question:

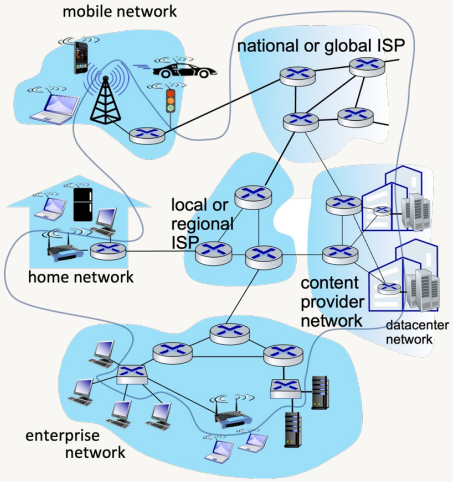
A new optical link doubles raw bit rate, but file-transfer completion time barely changes. Give one explanation from each of three networking areas.

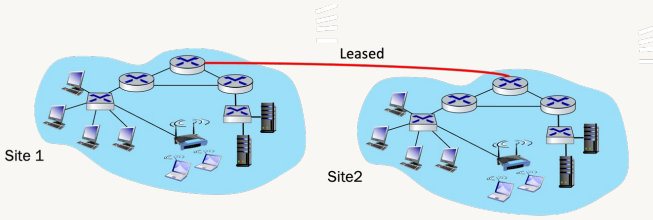
- Data communications: errors or signalling overhead limit usable rate
- Packet technology: a shared or slower access segment is the bottleneck
- TCP/IP/application: transport congestion control, server limits, latency, or application serialization prevents use of the extra capacity

Public and Private Parts of the Internet



- The Internet consists of networks owned and managed by different organizations
- Networks can be classified by ownership:
  - **Public networks:** Subscription-based services available through a service provider; “public” refers to availability, not data visibility
  - **Private networks:** Controlled by one organization with restricted access; they may use leased service-provider connections

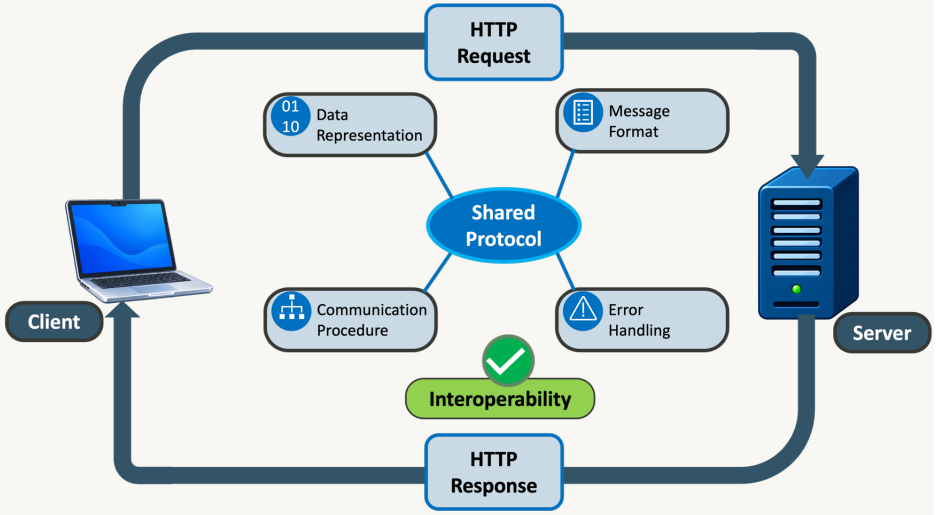




Networks, Interoperability, and Standards



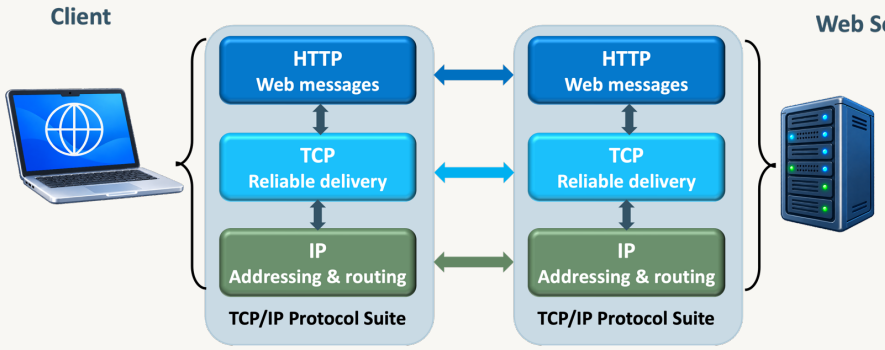
- Communication requires at least two entities: a sender and a receiver
- Both entities must agree on:
  - How data is represented
  - How communication begins and proceeds
  - How messages are formatted
- **Interoperability** requires entities to follow the same rules
- A **protocol** defines these rules, including error handling and responses to unexpected events

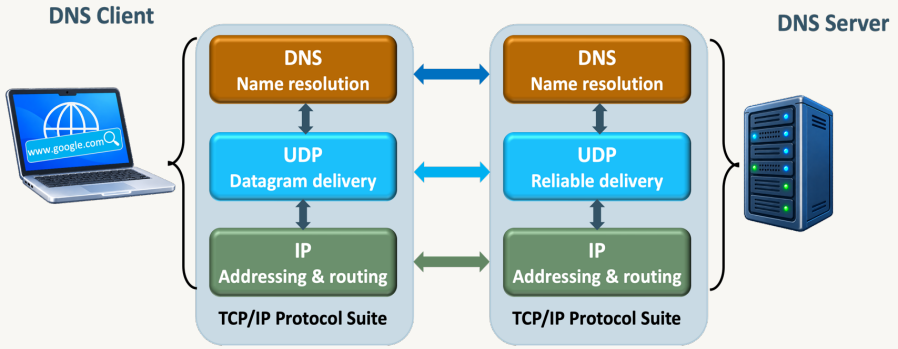


Protocol Suites and Layering Models



- A complete and efficient communication system requires a **set of protocols**
- Each protocol handles a specific networking function
- Protocols work together in **suites or families**
- Collectively, the suite covers all aspects of communication in a coordinated manner



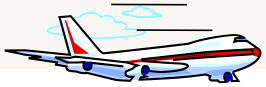


Real-World Example



Analogy:

- A traveler is transported by airline from source to destination
- A packet is transmitted from source host to destination host on the Internet



end-to-end transfer of person plus baggage

ticketing (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticketing (complaint)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

Why Layering?

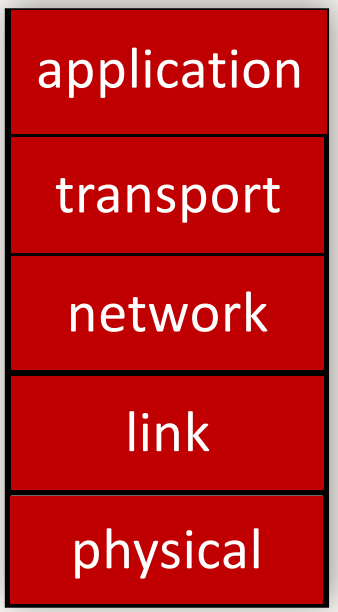


- Complex systems need a clear structure to define components and their relationships
- A **layered reference model** organizes the system
- **Modularization** simplifies maintenance and updates
  - Changes within one layer are transparent to other layers
  - Example: changing a gate procedure does not affect the rest of the system

Protocol Suites and Layering Models

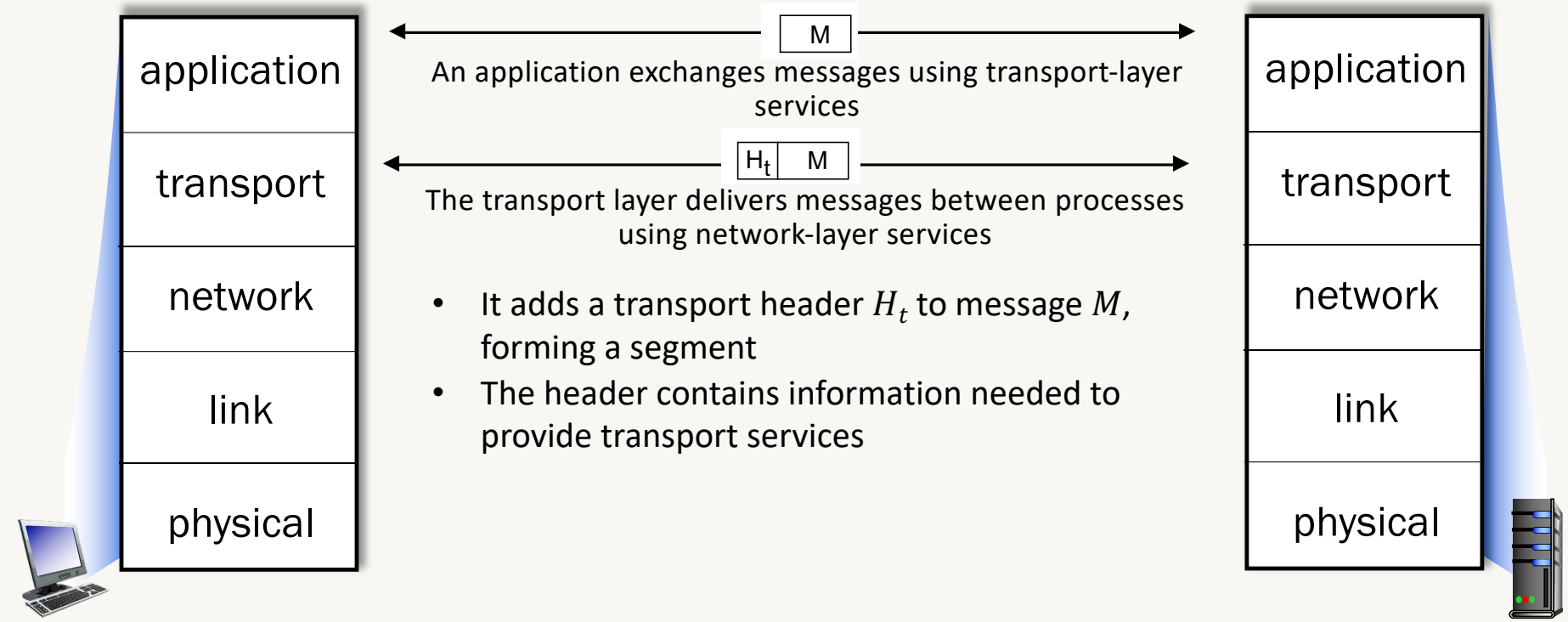


- **Application**: Supports network applications and defines message formats and meanings
  Examples: HTTP, IMAP, SMTP, DNS
- **Transport**: Provides process-to-process data transfer between applications
  Examples: TCP, UDP
- **Network**: Routes datagrams across interconnected networks
  Examples: IP, routing protocols
- **Link**: Transfers data between neighboring network devices
  Examples: Ethernet, Wi-Fi, PPP
- **Physical**: Transmits bits through the physical medium and hardware



Protocol Suites and Layering Models

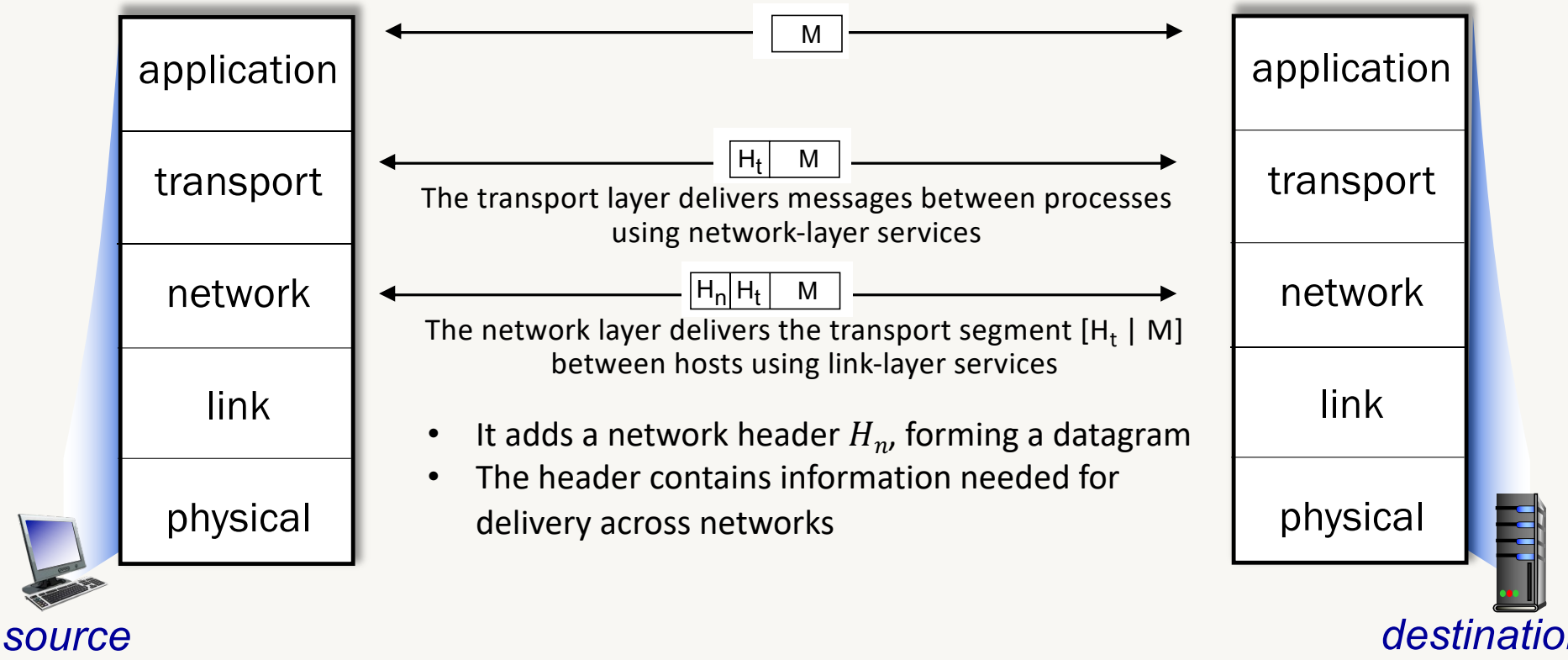




- It adds a transport header H_t to message M, forming a segment
- The header contains information needed to provide transport services

Protocol Suites and Layering Models

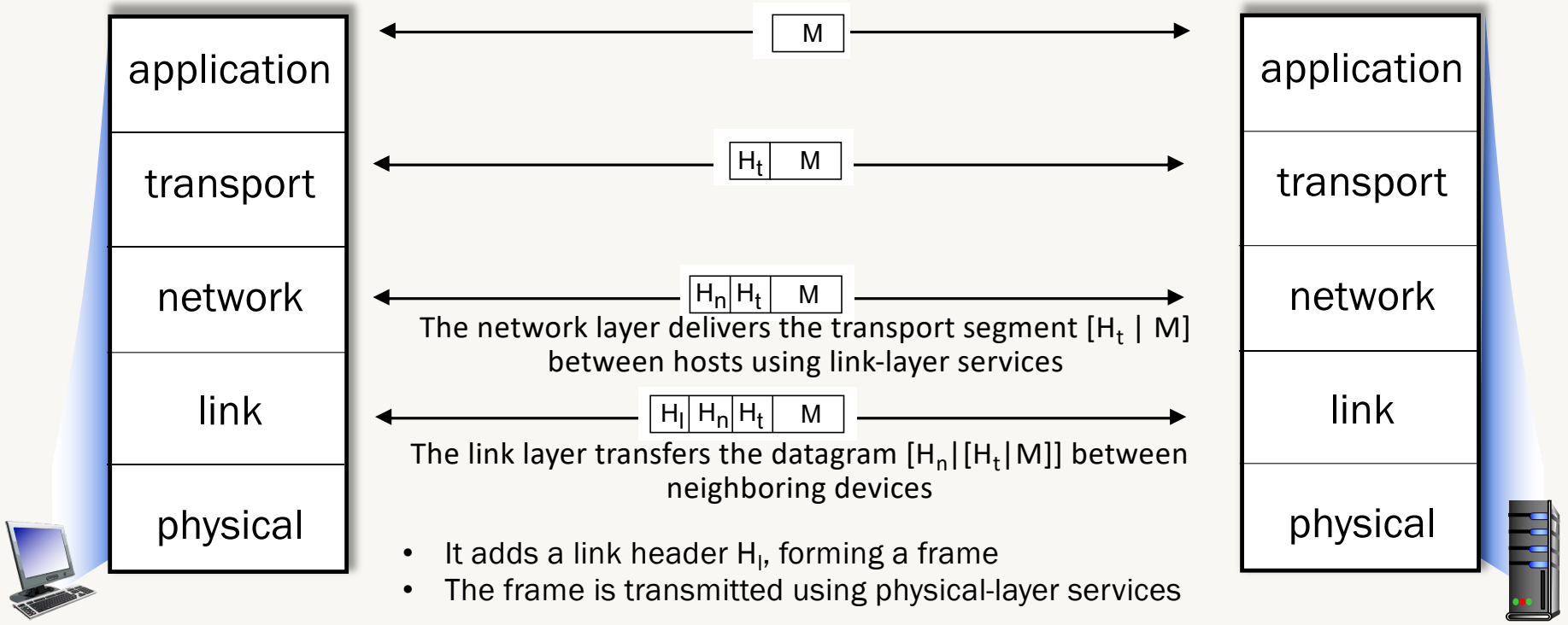




- It adds a network header H_n, forming a datagram
- The header contains information needed for delivery across networks

Protocol Suites and Layering Models





- It adds a link header H_l, forming a frame
- The frame is transmitted using physical-layer services

source *destination*



Discussion Question:

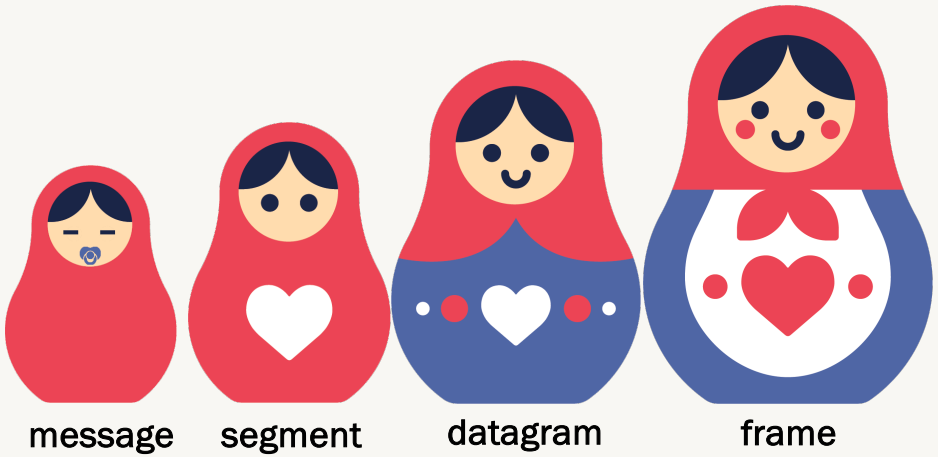
If a message is split into three transport segments, how many network and link headers are normally added on one link?



Encapsulation

Matryoshka dolls (stacking dolls)

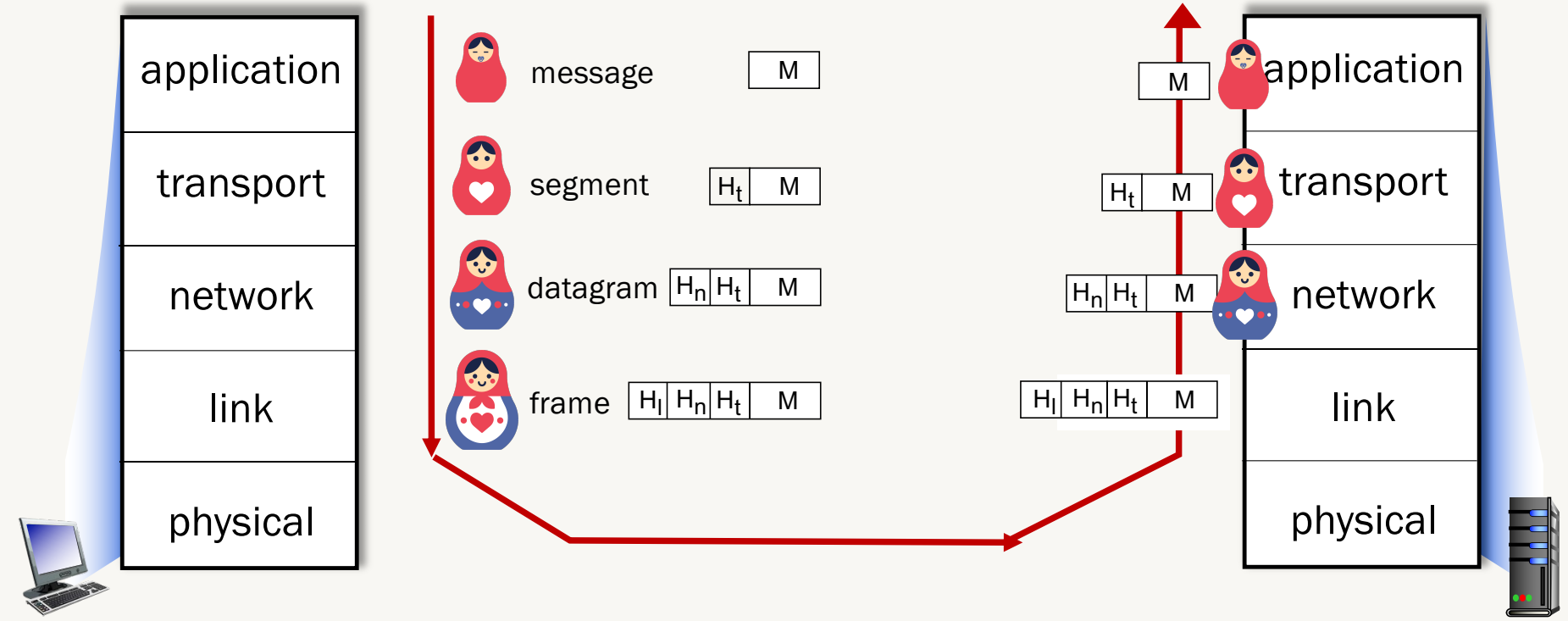




Credit: https://dribbble.com/shots/7182188-Babushka-Boi

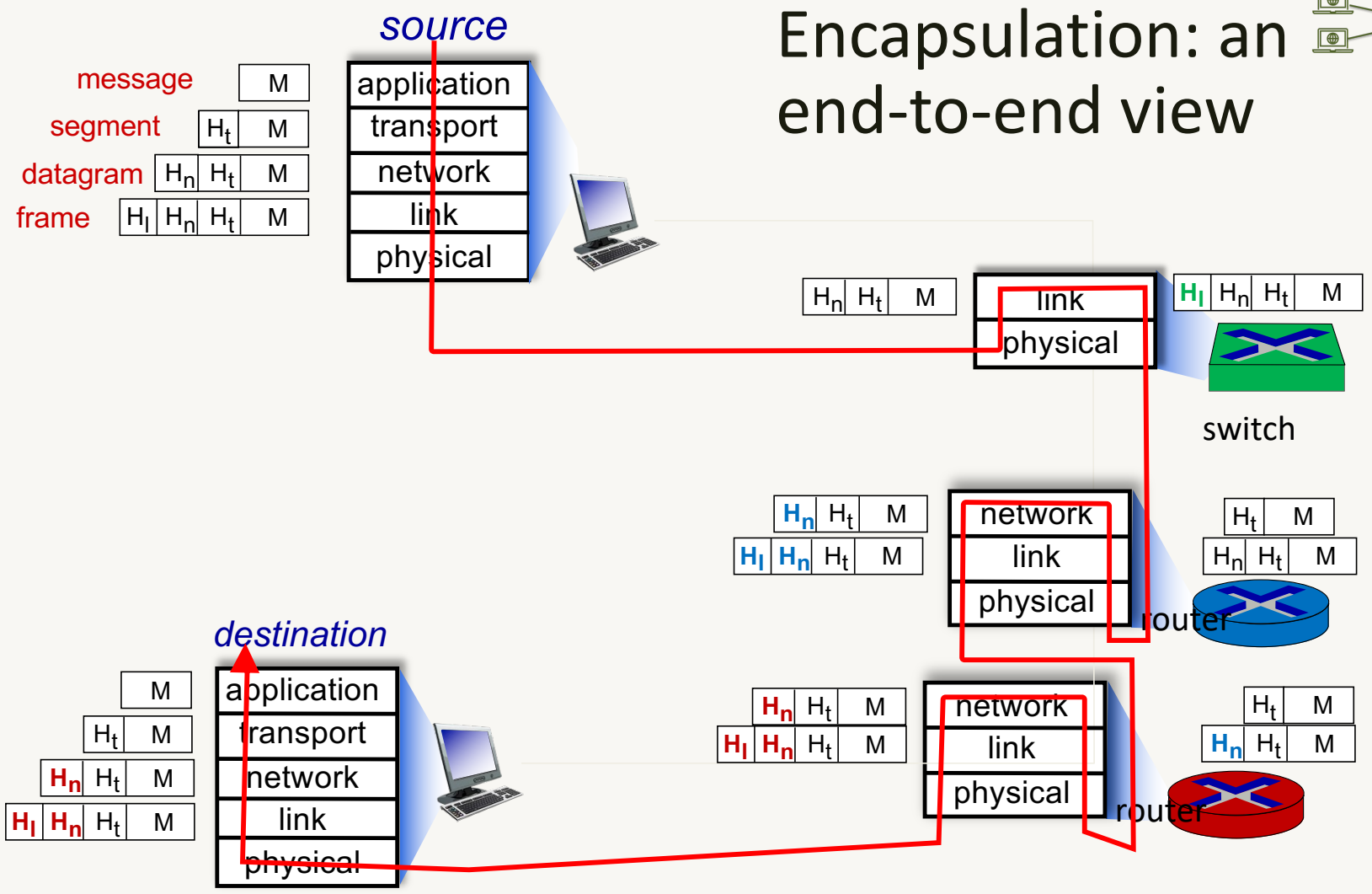
Protocol Suites and Layering Models





Encapsulation: an end-to-end view





References



- Ch. 1, Kurose and Ross
- Ch. 1, Comer



Questions?